Pituitary Disorders II: Approach to a Case of Hypopituitarism

Internal Medicine Department | Source: `10- approach to a case of hypopituitarism` (29 slides)

> **Legend:** ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.
>
> 🔗 **Bridge from Pituitary I:** that lecture covered **too much** hormone (prolactinoma, acromegaly) and **mass effects**. This one covers **too little**.
>
> - The same adenomas compress the normal gland and cause **hypopituitarism**.
> - Acute bleeding into them causes **pituitary apoplexy**.
> - The **portal (venous) blood supply** explains **Sheehan syndrome**.
> - **Thyroid 2** gave primary hypothyroidism (↑ TSH). Here we meet **secondary hypothyroidism** (↓ TSH).
> - **Hypoparathyroidism** gave **pseudohypoparathyroidism**, which reappears as a cause of **short stature**.

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Pathophysiology and etiology	**Sheehan** = commonest cause; why the anterior pituitary infarcts
Knowledge	⭐ Manifestations in children (dwarfism, Fröhlich) and adults (Simmonds)	2ry vs 1ry hypothyroidism; Simmonds vs Addison
Skill	⭐ Pituitary reserve tests	Insulin tolerance test; TRH; GnRH
Skill	⭐ Replacement order	**Steroid before thyroxine**
Skill	⭐ Emergency: **apoplexy**	Sudden headache + visual loss + shock → steroids

Lecture flow: Causes → Clinical picture (childhood: pituitary dwarfism, Fröhlich, Laurence–Moon–Biedl; adults: Simmonds / isolated deficiency) → 2ry vs 1ry hypothyroidism → Simmonds vs Addison → Investigations (GH, ACTH, TSH, LH/FSH reserve) → Treatment → **Pituitary apoplexy** → **Short stature** (causes, diagnosis, investigations, treatment) → Case.

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2. Main Content (Lecture Order)

2.1 Causes of Hypopituitarism (Slide 5) ⭐

#	Group	Causes
1	**Infarction**	⭐ **Postpartum necrosis (Sheehan syndrome): the MOST COMMON cause** (per the slide) after **postpartum haemorrhage**. Why the anterior pituitary is vulnerable: it **enlarges in pregnancy without more blood supply**, and its supply is **mainly venous (portal)**. Also **vascular disease**: vasculitis, DM

#	Group	Causes
2	**Infection / granuloma**	TB, syphilis, meningitis, **sarcoidosis**
3	**Idiopathic (congenital)**	Deficiency of one or all anterior hormones
4	**Neoplastic**	⭐ **Adenoma**, **craniopharyngioma**, leukaemia, lymphoma, metastasis
5	**Physical**	**Radiation, surgery, head trauma**
6	**Miscellaneous**	**Haemochromatosis**

[added] In modern series the commonest cause is a pituitary adenoma (or its treatment). Sheehan is the classic exam answer for postpartum hypopituitarism (failure of lactation + amenorrhoea after PPH). Mnemonic "8 I's":

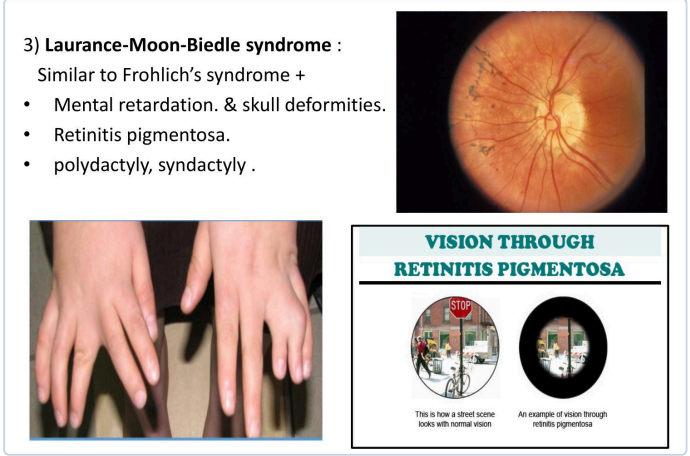
- *Invasive (tumour)*
- *Infarction (Sheehan, apoplexy)*
- *Infiltrative (sarcoid, haemochromatosis)*
- *Injury (trauma)*
- *Immunologic (lymphocytic hypophysitis)*
- *Iatrogenic (surgery, RT)*
- *Infectious*
- *Idiopathic*

2.2 Clinical Picture: Childhood (Slides 6–8)

Syndrome	Features
⭐ **Pituitary dwarfism (Levi–Lorain)**	**Proportionate** dwarfism, **childish facies**; later **hypogonadism** (± **cryptorchidism**). ↓ GH, commonly **idiopathic**; rarely inactive GH
⭐ **Fröhlich syndrome (dystrophia adiposogenitalis)**	**Dwarfism + hypogonadism** · **truncal and facial obesity sparing the extremities** ("sambousa-shaped") · **genu valgum** · **hypothalamic** disturbance: **DI, polyphagia, hypersomnia**
Laurence–Moon–Biedl	**Fröhlich features + mental retardation**, skull deformity, ⭐ **retinitis pigmentosa** (tunnel vision), **polydactyly / syndactyly**



Slide 7: hypopituitarism with gonadotrophin failure (39-year-old man); genu valgum



Slide 8: Laurence–Moon–Biedl (retinitis pigmentosa, polydactyly)

Slide 7 caption: extreme **genital atrophy**, **absence of body hair**, apparent **gynaecomastia** with obesity.

2.3 Clinical Picture: Adults (Slides 9–10) ⭐

- **Panhypopituitarism = Simmonds disease**, or **isolated hormone deficiency**.
- *[added]* Typical order of loss as the gland fails: **GH → LH/FSH → TSH → ACTH → PRL** ("**Go Look For The Adenoma Please**").

Hormone ↓	Result
Prolactin	⭐ **Failure of lactation** (the Sheehan clue)
GH	**Silent**; rarely hypoglycaemia
Gonadotrophins (post-pubertal)	Skin: **fine wrinkled**, **loss of secondary sexual hair** ("progeria"). Men: ↓ libido, impotence, testicular atrophy, oligospermia. Women: vaginal dryness, dyspareunia, **secondary amenorrhoea**, infertility, **breast atrophy**
TSH	**Secondary hypothyroidism**
ACTH	Usually not marked. **Hypoglycaemia**, **inability to withstand physical and psychological stress**, ↓ resistance to infection
Mass	Pressure neurological features (tumour)
⭐ **Coma**	**Hypothermia**, **hypoglycaemia**, or pressure on the brainstem/hypothalamus

Student pearl (slide 9): hypothalamic dopamine inhibits prolactin, whereas **TRH stimulates** it. *[added]* Hypothalamic or stalk lesions can therefore cause ↑ **PRL** together with other deficiencies.

2.4 Secondary vs Primary Hypothyroidism (Slide 11) ⭐

Feature	2ry (pituitary)	1ry (thyroid)
Severity	**Less severe**	More severe
Skin	**Fine, wrinkled**	More **oedematous** (myxoedema)
Hair	Dry, less coarse	Dry, **brittle, coarse**
Tongue	Not enlarged	**Enlarged**
Visual field defects	**May be present**	Absent
Heart / BP	**Small heart, ↓ BP**, visceral atrophy (**splanchnomicria**)	**Enlarged heart**, **HTN** common
Menses	**Amenorrhoea**	**Menorrhagia** (mostly)
Blood glucose	**Usually low**	Usually normal
Cholesterol	Normal	↑
⭐ **TSH**	**↓ (or inappropriately normal)**	↑
Other anterior hormones	↓	Normal

2.5 Simmonds Disease vs Addison's Disease (Slide 12) ⭐

Feature	Addison's (1ry adrenal)	Simmonds (pituitary)
Pigmentation	⭐ **Present** (↑ ACTH/POMC)	⭐ **Absent**: no lipotropin/MSH. Plus **pallor** (ACTH normally aids erythropoiesis) and **depigmented areolae** in women
Hypotension	**Marked** + hypovolaemia + **hyperkalaemia**	**Less marked**: **aldosterone continues** (renin–angiotensin control) → **no ↑ K^+ or hypovolaemia**
Bowel	Diarrhoea	**Constipation**
ACTH	↑	↓ (+ other pituitary hormones ↓)

Mnemonic: "Pale and Potassium-normal = Pituitary." Addison's is **dark and hyperkalaemic**.

🔁 **Recap: Causes and features.**

- **Causes:** Sheehan (enlarged gland in pregnancy, venous supply, PPH); vascular; infection or granuloma; congenital; tumours (adenoma, craniopharyngioma); surgery, RT, trauma; haemochromatosis.
- **Children:** proportionate dwarfism (Levi–Lorain); Fröhlich (dwarf, hypogonadal, truncal obesity, genu valgum, DI); LMB adds retinitis pigmentosa, polydactyly and MR.
- **Adults:** no lactation, silent GH loss, hypogonadism with fine wrinkled skin, secondary hypothyroidism (milder; small heart; low glucose; normal cholesterol), ACTH deficiency (pale, no hyperkalaemia, unlike Addison's). Coma from hypothermia or hypoglycaemia.

🔗 **Link:** Low hormone levels must be confirmed as low pituitary **reserve** with stimulation tests.

▶️ **Up next: Investigations and treatment.**

2.6 Investigations: Pituitary Reserve Tests (Slides 13–14) ⭐

Alongside imaging of any tumour:

Axis	Test	Normal response
GH	Screen: GH during **sleep** and after **exercise** (should rise **> 6 ng/mL**). If not: stimulation tests. ① **insulin-induced hypoglycaemia** (0.1 U/kg IV) ② **L-dopa** 250 mg PO ③ **clonidine** (50 µg per 10 years of age above 5, max 250 µg PO). Samples **fasting, 60 and 90 min**	**GH > 6 ng/mL excludes GH deficiency**
⭐ **ACTH**	**Insulin-induced hypoglycaemia** (insulin tolerance test)	**Cortisol ↑ > 10 µg/dL above basal** = normal
TSH	**TRH test**: 500 µg IV TRH	**Doubling of basal TSH**. **Low response** in **2ry hypothyroidism** (and in **thyrotoxicosis**)
LH/FSH	A. **Normal cyclic menses or normal sperm count EXCLUDES hypothalamic–pituitary disease** · B. basal LH/FSH · C. **GnRH (LHRH) test** 100 µg IV · D. **clomiphene** test (oestrogen antagonist)	C and D: **LH doubles, FSH ↑ ~50%**

[added] Insulin tolerance test is the gold standard for GH and ACTH reserve. It is contraindicated in IHD, epilepsy and the elderly. Practical first steps: 9 am cortisol, free T4 with TSH (low T4 + low/normal TSH = central), LH/FSH with testosterone/oestradiol, IGF-1, prolactin, Na^+, MRI.

2.7 Treatment (Slides 15–17) ⭐

	Detail
A. Surgery	For a **space-occupying lesion** (trans-frontal or trans-sphenoidal)
B. Radiotherapy	**Incomplete surgical removal**; **non-surgical candidates**
C. Hormone replacement	
1. ⭐ **Glucocorticoid**	e.g. **prednisone 5 mg morning + 2.5 mg noon** orally (*[added]* or hydrocortisone 15–25 mg/day; **double in illness ("sick-day rules")**)

	Detail
2. ⭐ **L-thyroxine**	**100–250 µg/day**. ⭐ **Give AFTER cortisol replacement** to avoid precipitating **adrenal crisis**
3. **GH**	**3 injections/week before epiphyseal closure**
4. **Gonadotrophins / sex steroids**	**Fertility not needed**: men → **long-acting IM testosterone**; women → **cyclic oestrogen + progesterone**. **Fertility needed**: **hMG** (FSH activity) + **hCG** (LH activity) for several months
5. **ADH**	**Desmopressin** intranasally or pitressin tannate IM

[added] No mineralocorticoid is needed (aldosterone is preserved). In secondary hypothyroidism, monitor free T4, not TSH.

Clinical pearls (slide 17):

- **Pituitary incidentalomas** → evaluate for **hypopituitarism and hormonal hypersecretion**.
- **Caution:** about **one-third** of pituitary adenomas have **associated hypopituitarism**, so screen all patients with a pituitary mass.
- **Caution:** adenomas can bleed (**apoplexy**), causing **severe headache, visual symptoms, CV collapse and/or secondary adrenal insufficiency**.

> 🔁 **Recap: Tests and treatment.**
> - **Reserve tests:** GH (sleep/exercise, then insulin/L-dopa/clonidine; > 6 normal); ACTH (insulin hypoglycaemia; cortisol ↑ > 10); TSH (TRH should double TSH); LH/FSH (normal menses or sperm excludes disease; GnRH/clomiphene double LH).
> - **Treatment:** surgery or RT for a mass. Replace **steroid FIRST, then thyroxine**, plus GH (children), sex steroids or hMG/hCG for fertility, desmopressin for DI.
>
> 🔗 **Link:** Sometimes the gland fails in minutes rather than years.
> ▶️ **Up next: Pituitary apoplexy.**

2.8 Pituitary Apoplexy (Slides 18–19) ⭐

Causes:

1. **Haemorrhage into a pituitary tumour** (*[added]* often an undiagnosed macroadenoma; triggers: anticoagulants, dopamine agonists, surgery, pregnancy)
2. **CVA** (particularly in **diabetics**)
3. **Sickle cell anaemia**

Clinical picture:

1. ⭐ **Sudden headache**, often with **rapidly worsening visual field defects or diplopia** (compression of surrounding nerves).
2. Then **acute hormone deficiency**, predominantly **adrenal insufficiency** (hypotension, shock). It rarely causes **auto-hypophysectomy**, which **cures** acromegaly or Cushing's.
3. **↑ ICP**.
4. **Meningeal irritation**.
5. **Confusion → coma** (with hyperpyrexia). **Fatal if not treated urgently.**

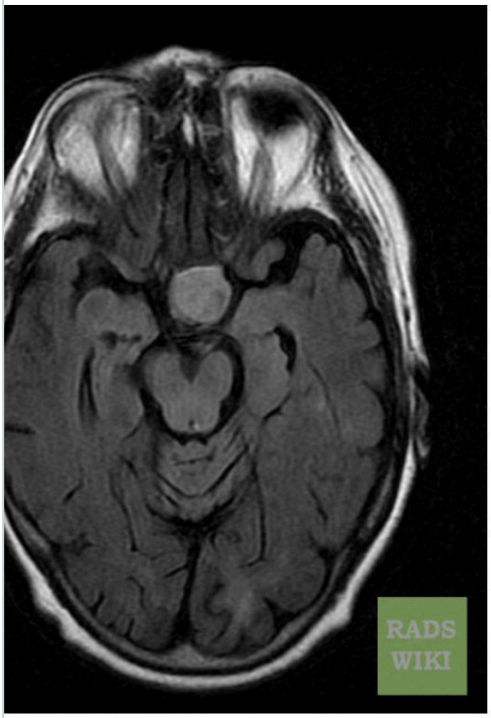
Clinical pearl: apoplexy, from acute hypocortisolism and hypothyroidism, can present with **severe headache, sudden hypotension and hypovolaemic shock**.

Investigations:

1. ⭐ **CT scan is diagnostic** (*[added]* MRI is more sensitive).
2. **CSF bloody** (*[added]* mimics SAH).

Treatment:

1. ⭐ **Glucocorticoids** (e.g., **dexamethasone**; *[added]* high-dose hydrocortisone 100 mg IV is standard)
2. **Surgical decompression** if needed (*[added]* for deteriorating vision or consciousness)



Slide 19: haemorrhagic pituitary on imaging

> 🔁 **Recap: Apoplexy.** Bleeding or infarction of an adenoma (or in DM or sickle cell). Thunderclap headache + visual loss or ophthalmoplegia + shock (acute ACTH loss) ± meningism, coma. CT/MRI shows it; CSF may be bloody. Give steroids immediately, then decompress if vision or consciousness worsens.
>
> 🔗 **Link:** In children, the chronic version of pituitary failure presents as growth failure, but most short children have other causes.
>
> ▶️ **Up next: Short stature.**

2.9 Short Stature (Slides 20–24) ⭐

Causes:

Primary: intrinsic to the skeleton	**Secondary: outside the skeleton**
Skeletal dysplasia (chondrodystrophy, e.g. **achondroplasia**)	**IUGR** (infection, placental insufficiency, maternal illness)
Chromosomal: **Down** (trisomy 21), ⭐ **Turner**	⭐ **Idiopathic constitutional growth delay: the MOST COMMON cause**

Primary: intrinsic to the skeleton	Secondary: outside the skeleton
Inborn errors: glycogen storage disease, mucopolysaccharidosis	**Malnutrition** (kwashiorkor, marasmus)
Genetic short stature (e.g. pygmies)	**GIT**: **coeliac**, cystic fibrosis, chronic liver disease, IBD
	Renal: chronic parenchymal disease, **RTA**
	Cardiorespiratory: congenital heart disease, asthma
	Endocrine: **hypopituitarism / GH or GHRH ↓** (or GH resistance), **pseudohypoparathyroidism**, **precocious puberty** (sex-steroid excess fuses epiphyses), **Cushing's**, **cretinism / juvenile myxoedema** (4–12 years)
	Psychological deprivation

Diagnosis (slide 22):

- **Height within 2 SD** of the mean may be normal. **< 3 SD → full evaluation.**
- **Body proportions: short limbs vs trunk** → long-standing **hypothyroidism** or **achondroplasia**.
- **Height-to-weight ratio** (*[added]* **underweight for height** suggests GI or nutritional disease; **overweight for height** suggests endocrine disease).
- **Specific findings:**

Condition	Clue
Achondroplasia	**Macrocephaly**, short limbs
Cretinism	Dry skin, coarse hair, **depressed nasal bridge**, big lips, **large protruding tongue**
Cushing's	Central obesity, striae, HTN
⭐ **Turner**	**Webbed neck, shield chest, coarctation**, delayed puberty
GH ↓	Proportionate, childish facies
Pseudohypoparathyroidism	**Moon face, obesity, short metacarpals**, mental retardation

Investigations (slide 23):

1. **Hormones:** thyroxine, cortisol, GH (and GH response to exercise/clonidine).
2. **Renal:** urea, creatinine.
3. **Malabsorption** and CVS work-up.
4. ⭐ **Karyotype** (gonadal dysgenesis: **Turner**).
5. **Ca** (↓ in **pseudohypoparathyroidism**), **P** (↓ in **vit D-resistant rickets**), **HCO_3** (↓ in **RTA**).
6. **Skull X-ray** for CNS lesions (e.g., **craniopharyngioma**: suprasellar calcification *[added]*).
7. ⭐ **Bone age (hand X-ray): DELAYED in secondary defects, NOT retarded in primary defects.**

Treatment (slide 24):

1. **Treat the cause early** (Cushing's, hypothyroidism…).
2. **GH deficiency → recombinant GH 0.1 U/kg SC 3×/week.**

NB: in **primary** defects, bone-growth potential is **severely impaired**, unlike secondary defects.

> 🔁 **Recap: Short stature.**
>
> - **Primary (skeletal) causes:** achondroplasia, Turner, Down, inborn errors, genetic. Bone age is NOT delayed and growth potential is poor.
> - **Secondary causes:** constitutional delay (the commonest), IUGR, malnutrition, coeliac, IBD, CF, renal/RTA, CHD, endocrine (GH ↓, hypothyroid, Cushing's, PHP, precocious puberty), deprivation. Bone age is delayed and catch-up is possible.
> - **Evaluate** if < 3 SD: proportions, weight-for-height, stigmata; TFT, cortisol, GH stimulation, renal, coeliac, karyotype, Ca/P/HCO_3, skull X-ray, bone age.
> - **Treatment:** treat the cause; GH if deficient.

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3. High-Yield Comparison Tables

Table A: Primary vs Secondary Hypothyroidism *(slide 11)*

	2ry	1ry
TSH	**↓ / N**	**↑**
Other pituitary hormones	↓	N
Cholesterol	**N**	↑
Glucose	**Low**	N
Heart / BP	**Small, ↓BP**	Big, HTN
Menses	**Amenorrhoea**	Menorrhagia
Tongue / skin	N / fine wrinkled	**Big** / oedematous
Visual fields	**May be ↓**	N

Table B: Simmonds vs Addison *(slide 12)*

	Simmonds	Addison
Skin	**Pale**, depigmented areolae	**Pigmented**
ACTH	**↓**	**↑**
K^+ / volume	**Normal** (aldosterone intact)	**↑ K^+**, hypovolaemia
BP	Mildly ↓	Markedly ↓
Bowels	Constipation	Diarrhoea
Mineralocorticoid needed?	**No** *[added]*	**Yes** (fludrocortisone) *[added]*

Table C: Pituitary Reserve Tests

Axis	Stimulus	Normal
GH	Sleep/exercise → **insulin, L-dopa, clonidine**	GH > **6 ng/mL**
ACTH	**Insulin hypoglycaemia**	Cortisol ↑ > **10 µg/dL**
TSH	**TRH 500 µg IV**	TSH **doubles**
LH/FSH	**GnRH 100 µg IV** / **clomiphene**	LH **doubles**, FSH ↑ 50%

Table D: Childhood Hypopituitary Syndromes

	Levi–Lorain	Fröhlich	Laurence–Moon–Biedl
Dwarfism	**Proportionate**	✅	✅
Obesity	—	**Truncal**, spares limbs	✅
Hypogonadism	Later	✅	✅
Extra	Childish facies	**Genu valgum, DI, polyphagia, hypersomnia**	**Retinitis pigmentosa, polydactyly, MR**, skull deformity

Table E: Short Stature, Primary vs Secondary

	Primary (skeletal)	Secondary (non-skeletal)
Examples	Achondroplasia, Turner, Down, IEM, genetic	**Constitutional delay**, malnutrition, chronic disease, endocrine
Bone age	**Not delayed**	**Delayed**
Growth potential	Severely impaired	Preserved (catch-up)

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4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Replace deficient hormones	⭐ **Glucocorticoid BEFORE thyroxine**: thyroxine first can precipitate adrenal crisis
Adrenal insufficiency → ↑ K^+	**Secondary (pituitary) → no ↑ K^+** (aldosterone preserved) and **no pigmentation**
Hypothyroidism → ↑ TSH	**Secondary → low/normal TSH**: measure **free T4**
Low TRH response = pituitary failure	**Also in thyrotoxicosis** (suppressed TSH)
GH deficiency in adults causes symptoms	It is **silent** (rare hypoglycaemia)
Lactation failure = breastfeeding problem	After PPH = **Sheehan**
Short = endocrine disease	**Constitutional delay is the commonest**
Treat short stature with GH	Only if **GH deficient**. Primary skeletal defects respond poorly
Apoplexy = tumour bleed	Also **DM (CVA)** and **sickle cell**. It can rarely **cure** acromegaly or Cushing's
Normal menses tell you little	Normal cyclic menses or sperm count **excludes** hypothalamic–pituitary disease (slide 14)

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5. Clinical Correlations

- **Woman with massive PPH who can't breastfeed, then amenorrhoea, cold intolerance, pallor, loss of axillary hair** → **Sheehan syndrome** → steroid then thyroxine; oestrogen/progesterone.

- **Known macroadenoma, sudden thunderclap headache, ophthalmoplegia, BP 85/50** → **apoplexy** → **IV hydrocortisone**, MRI, neurosurgery.
- **Pale, non-pigmented patient with hyponatraemia and normal K^+** → **secondary adrenal insufficiency** (pituitary), not Addison's.
- **Patient started on levothyroxine for "hypothyroidism" with low TSH, who then collapses** → unmasked **ACTH deficiency**. Give steroid first.
- **Obese boy with small genitals, genu valgum and polyuria** → **Fröhlich** (hypothalamic lesion, e.g. craniopharyngioma).
- **Child with night blindness, polydactyly, obesity, hypogonadism** → **Laurence–Moon–Biedl (Bardet–Biedl)**.
- **Short girl with webbed neck, coarctation, primary amenorrhoea** → **Turner (45 XO)**: karyotype, GH, oestrogen.
- **Short child with chronic diarrhoea, iron deficiency, underweight for height** → **coeliac disease**.
- **Child with short stature and suprasellar calcification on skull X-ray, visual field defect** → **craniopharyngioma**.

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6. Rapid Revision (Last-Day)

1. Causes: **Sheehan (commonest per slide)**; vascular (DM, vasculitis); TB, syphilis, meningitis, **sarcoid**; congenital; **adenoma, craniopharyngioma**, metastasis, leukaemia/lymphoma; **RT, surgery, trauma**; **haemochromatosis**.
2. **Sheehan:** the anterior pituitary **enlarges in pregnancy without more blood supply** and its supply is **venous** → infarction after PPH.
3. Childhood: **Levi–Lorain** (proportionate dwarfism, childish face, later hypogonadism ± cryptorchidism); **Fröhlich** (dwarfism, hypogonadism, **truncal obesity**, **genu valgum**, DI, polyphagia, hypersomnia); **LMB** (+ **retinitis pigmentosa**, **polydactyly**, MR).
4. Adults = **Simmonds**: **↓PRL → no lactation**; GH silent; gonadotrophins → fine wrinkled skin, loss of sexual hair, amenorrhoea, impotence, breast/testicular atrophy; **2ry hypothyroidism**; ACTH → hypoglycaemia, poor stress tolerance, infection; **coma** from hypothermia or hypoglycaemia.
5. **2ry vs 1ry hypothyroidism:** milder, fine skin, no big tongue, **small heart/low BP**, **amenorrhoea**, **low glucose**, **normal cholesterol**, **↓ TSH**.
6. **Simmonds vs Addison: pale** vs pigmented; **no hyperK/hypovolaemia** vs present; **constipation** vs diarrhoea; **ACTH ↓** vs ↑.
7. GH reserve: sleep/exercise GH **> 6**; else **insulin 0.1 U/kg, L-dopa 250 mg, clonidine** (0, 60, 90 min).
8. ACTH reserve: **insulin hypoglycaemia → cortisol ↑ > 10 µg/dL**.
9. TSH: **TRH 500 µg** doubles TSH; low in **2ry hypothyroidism and thyrotoxicosis**.
10. LH/FSH: **normal menses or sperm count excludes disease**; GnRH 100 µg / clomiphene → **LH doubles, FSH +50%**.
11. Treatment: surgery/RT for a mass; **prednisone 5 mg am + 2.5 mg noon FIRST**, **then L-T4 100–250 µg**; GH 3×/week (children); testosterone IM or cyclic E+P; **hMG + hCG for fertility**; **desmopressin** for DI.
12. ⅓ of adenomas have hypopituitarism; screen all.
13. **Apoplexy:** tumour bleed, CVA in DM, sickle cell. **Sudden headache + visual loss/diplopia + adrenal crisis**, ↑ ICP, meningism, coma. **CT diagnostic, bloody CSF**. **Glucocorticoids + decompression.**
14. Short stature: **constitutional delay most common**. Primary (achondroplasia, Turner, Down, IEM) vs secondary (IUGR, malnutrition, coeliac, CF, IBD, liver, renal/RTA, CHD, asthma, endocrine, deprivation).

15. Evaluate if **< 3 SD**. Short limbs → hypothyroidism or achondroplasia. **Bone age delayed in secondary, not in primary.**
16. GH deficiency → **rhGH 0.1 U/kg SC 3×/week**.

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7. Exam-Style Questions

Q1 (lecture case, slides 25–26). A 39-year-old woman with a history of migraine has a **sudden severe throbbing headache and double vision for 1 hour**. She has had frequent headaches and **amenorrhoea for months**. BP **93/61**. Visual fields: **bilateral lateral-quadrant (bitemporal) defect**. Most likely diagnosis? a) TIA · b) **Pituitary apoplexy** · c) Sheehan syndrome · d) Migraine with aura · e) SAH · f) Cluster headache

> **Answer: b. Pituitary apoplexy.** *[reasoning added; key not on slides]*
> - The **prodrome** (headaches + amenorrhoea) suggests an underlying **macroadenoma, likely a prolactinoma**.
> - Then: **sudden headache + diplopia** (cavernous sinus), **bitemporal field loss** (chiasm) and **hypotension** (acute ACTH loss).
> - **SAH** gives no field defect or hypotension from adrenal failure.
> - **Sheehan** needs a postpartum haemorrhage.
> - **Migraine with aura** doesn't cause hypotension or a fixed bitemporal defect.

Q2. Six months after a delivery complicated by massive haemorrhage, a woman reports **failure of lactation**, amenorrhoea, fatigue and cold intolerance. Labs: low free T4, **low TSH**, Na 129, K 4.2. Which treatment should be started FIRST? A. Levothyroxine · B. **Hydrocortisone (glucocorticoid)** · C. Fludrocortisone · D. Oestrogen–progesterone · E. Cabergoline

> **Answer: B.**
> - This is **Sheehan syndrome**. ⭐ **Steroid before thyroxine** to avoid adrenal crisis (slide 15).
> - **Fludrocortisone is unnecessary**: aldosterone is intact and K^+ is normal.

Q3. Which feature favours **pituitary (secondary)** over **primary** adrenal insufficiency? A. Hyperpigmentation · B. Hyperkalaemia · C. **Pallor with depigmented areolae** · D. Marked hypovolaemia · E. Raised ACTH

> **Answer: C** (slide 12).

Q4. A 10-year-old's height is −3.5 SD with **delayed bone age**, proportionate body, normal weight for height, low IGF-1 and a **GH peak of 3 ng/mL after clonidine**. What is the treatment? A. Observe; constitutional delay · B. **Recombinant GH 0.1 U/kg SC 3×/week** · C. Levothyroxine · D. Oestrogen · E. Nothing; primary skeletal defect

> **Answer: B.**
> - **GH deficiency** (stimulated peak < 6 ng/mL) → rhGH (slides 13, 24).
> - Delayed bone age means a secondary defect with growth potential.

Q5. The TRH test produces a **blunted TSH rise** in which conditions? A. Primary hypothyroidism only · B. **Secondary hypothyroidism and thyrotoxicosis** · C. Euthyroid sick only · D. Iodine deficiency · E. Hashimoto

Answer: B (slide 14).

- In thyrotoxicosis, high T4 suppresses the thyrotrophs.
- In primary hypothyroidism the response is **exaggerated**.

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8. Exam Pitfalls & Traps

- **Thyroxine before steroid** → adrenal crisis. Always **steroid first**.
- **"Most common cause" conflict:** the slide says **Sheehan**. Modern data: **pituitary adenoma** *[added]*. Answer by the stem: postpartum → Sheehan; general adult → adenoma.
- **Secondary adrenal insufficiency: no** hyperkalaemia, **no** pigmentation, **no** fludrocortisone needed. Hyponatraemia can still occur (cortisol is needed to excrete free water) *[added]*.
- **Secondary hypothyroidism: TSH can be normal**. **Free T4 is the test and the treatment target.**
- **Normal menses or sperm count** rules out hypothalamic–pituitary disease (slide 14), a quick bedside reserve check.
- **GH test thresholds:** > 6 ng/mL = normal (lecture). *[added]* Many modern labs use lower cut-offs (e.g. > 5–10 µg/L depending on assay). Use the lecture number.
- **Insulin tolerance test** is contraindicated in **IHD and epilepsy** *[added]*.
- **Apoplexy imaging:** the slide says **CT is diagnostic**. MRI is more sensitive *[added]*. **Bloody CSF** can mislead you into diagnosing SAH.
- **Bone age: delayed in secondary** causes (endocrine, chronic disease, constitutional), **not delayed in primary skeletal** disorders.
- **Precocious puberty** gives an early tall child but a **short adult** (early epiphyseal fusion).
- **Slide 6** "samboxa-shaped obesity" = truncal obesity sparing the limbs (a local food analogy).